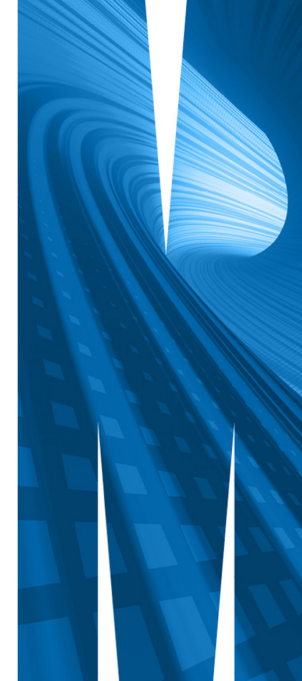
MONASH University

ETX1100/ETX5900 Business Statistics

Week 5: Chi-square and simple regression

5 January 2027



Outline

1. What does “related” mean?
2. Testing categorical independence
3. Numerical relationships
4. Simple linear regression
5. Validation and responsible prediction
6. Integrated application

Today's journey

1. Test independence between categorical variables.
2. Visualise and quantify numerical relationships.
3. Distinguish association from causation.
4. Build and interpret a simple linear regression in Excel.
5. Predict responsibly and diagnose limitations.

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Retrieval check

- How would you assess association between two categorical variables?
- What does independence mean in probability language?
- What does a small p-value say—and not say?
- Which variable types can be placed on a scatterplot?

Match the method to the variables

Explanatory variable	Outcome variable	Useful tools
categorical	categorical	contingency table, conditional probability, chi-square test of independence
numerical	numerical	scatterplot, covariance, correlation, regression

Categorical association

Two categorical variables are associated when the outcome distribution changes across explanatory-variable groups.

Compare conditional probabilities such as:

$$P(\text{tried} \mid \text{high income})$$

with the corresponding probabilities in other income groups or with the marginal probability.

Watch me: categorical relationship

Using a contingency table, ask:

1. What is the outcome?
2. What groups are being compared?
3. Are row or column percentages appropriate?
4. Do conditional probabilities differ meaningfully?
5. Is causal language justified?

Now you try: explain the association

Time allowed: 5 minutes

A loyalty-app sample finds 72% high satisfaction among users and 61% among non-users.

1. State the observed association.
2. Explain why the two percentages are conditional.
3. Give two plausible confounders.
4. Rewrite “the app increases satisfaction” as a defensible conclusion.

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From percentages to a population claim

Different sample percentages suggest association. A **chi-square test of independence** asks whether the differences are large enough, relative to sampling variation, to support an association in the population.

- H_0: the two categorical variables are independent in the population.
- H_1: they are associated in the population.

Use counts from a random or otherwise defensible sample of independent observations. Each person or unit belongs in one cell only.

Expected counts under independence

For row i and column j, the count expected under H_0 is

$$E_{ij} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{n}.$$

Compare **observed counts** O_{ij} with these expected counts. A large difference is evidence against independence. Check that all expected counts are at least 5 for the approximation used here.

Watch me: support channel and resolution

Open `week5-chi-square-data.xlsx`, sheet **Service**. Each row is one customer case.

1. PivotTable: `Channel` in rows, `Resolved` in columns, count `Case ID`.
2. Copy the four counts to a range and add row and column totals.
3. Compare within-channel resolution rates; calculate expected counts.
4. Check that every expected count is at least 5.

Now you try: expected counts

Time allowed: 6 minutes

Using the **Service** table:

1. State H_0 and H_1 in context.
2. Calculate the expected count for online cases that were resolved.
3. Complete the other three expected counts in Excel.
4. Explain why row or column totals must not be included in the test ranges.

Test statistic

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}, \qquad df = (r-1)(c-1).$$

For each cell, calculate `=(observed-expected)^2/expected`; sum the contributions. The result is always non-negative and grows as observed counts move away from expected counts.

Excel p-value and conclusion

Use `=CHISQ.TEST(observed_range,expected_range)` for the right-tail p-value. The ranges must have the same shape and exclude totals.

At 5%, reject H_0 when $p < 0.05$. Otherwise, the data do not give sufficient evidence of association. Do not claim independence has been proved.

Watch me: audit the Excel result

For the **Service** table, check that `=CHISQ.DIST.RT(test_statistic,df)` agrees with `=CHISQ.TEST(observed_range,expected_range)`. The sum of contributions gives the test statistic.

Ask whether the result is statistically persuasive **and** whether the observed percentage gap matters for the service decision. Neither a small p-value nor this observational table establishes that the channel caused resolution.

Now you try: loyalty and renewal

Time allowed: 10 minutes

On **Loyalty**, count `Renewed` within each `Tier` using a PivotTable.

1. Compare renewal percentages by tier and state the hypotheses.
2. Build expected counts and check the minimum.
3. Calculate df, χ^2 and the `CHISQ.TEST` p-value.
4. Conclude at 5% and name one design limitation.

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Scatterplots first

A scatterplot reveals:

- **direction:** positive or negative;
- **form:** linear or nonlinear;
- **strength:** tight or diffuse pattern; and
- **unusual observations:** outliers or influential points.

Put the explanatory variable X on the horizontal axis and response Y on the vertical axis.

Covariance

Sample covariance measures joint movement:

$$s_{XY} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}.$$

- Positive: variables tend to move in the same direction.
- Negative: variables tend to move in opposite directions.
- Magnitude depends on measurement units, limiting comparison.

Correlation

Correlation standardises covariance:

$$r = \frac{s_{XY}}{s_X s_Y}, \qquad -1 \leq r \leq 1.$$

- Sign gives direction.
- Magnitude gives strength of **linear** association.
- Correlation is unit-free and symmetric in X and Y.

Excel: `=CORREL(x_range,y_range)`.

Watch me: income and spending

Open `week5-regression-data.xlsx`, sheet **IncomeSpend**.

1. Insert a scatterplot of `Spend` against `Income`.
2. Label axes with units.
3. Add a linear trendline.
4. Calculate `=CORREL(Income,Spend)`.
5. Describe direction, form, strength, and unusual points.

Now you try: age and market price

Time allowed: 7 minutes

On **HouseSize**:

1. plot market price against house age;
2. add a linear trendline;
3. calculate correlation;
4. interpret its sign and magnitude; and
5. identify whether the plot supports a linear summary.

Correlation cautions

- $r \approx 0$ does not rule out a nonlinear relationship.
- One influential observation can alter r greatly.
- Restricted ranges can weaken an observed correlation.
- Combining groups can create a misleading aggregate pattern.
- Correlation does not establish causation.

Now you try: correlation critique

Time allowed: 5 minutes

For each statement, identify the problem:

1. "$r = 0$, so the variables are unrelated."
2. "Sales and advertising are correlated, so advertising caused every sale."
3. "The correlation is 0.8 kilograms."
4. "Removing the unusual store is fine because it improves r."

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Population and fitted models

Population model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

Fitted sample model:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

Residual:

$$e_i = Y_i - \hat{Y}_i$$

Least squares

Least squares chooses the line that minimises:

$$SSE = \sum_{i=1}^{n} e_i^2.$$

Squaring prevents positive and negative residuals cancelling and penalises larger errors more strongly.

The fitted line describes the mean response predicted at each X.

Interpreting coefficients

- **Intercept $\hat{\beta}_0$:** predicted Y when $X = 0$.
- **Slope $\hat{\beta}_1$:** predicted mean change in Y for a one-unit increase in X.

Include both variables, units, direction, and “on average”.
Question whether $X = 0$ is observed or meaningful.

Watch me: regression in Excel

For `Spend` as Y and `Income` as X:

1. `Data` → `Data Analysis` → `Regression`;
2. select the labelled Y and X ranges;
3. tick **Labels** and request residual output;
4. locate coefficients, R^2, standard errors, and p-values; and
5. write the fitted equation.

Now you try: interpret IncomeSpend

Time allowed: 7 minutes

From the Excel output:

1. report and interpret the intercept;
2. report and interpret the slope;
3. predict the spending change for an AUD 10,000 income increase;
4. state whether the intercept is practically meaningful; and
5. compare the slope sign with the correlation sign.

Coefficient test

To test for a linear relationship:

$$H_0 : \beta_1 = 0, \qquad H_1 : \beta_1 \neq 0.$$

Excel reports the two-sided p-value for the slope. A small p-value supports a population linear association, subject to model assumptions and study design.

Coefficient of determination

$$R^2 = \frac{\text{variation explained by the model}}{\text{total variation in } Y}.$$

Interpret R^2 as the percentage of sample variation in the **response** explained by its linear relationship with X.

It is not the percentage of observations predicted correctly.

Watch me: predict from the line

For the fitted model

$$\widehat{Spend} = 8254.575 + 0.296(Income),$$

substitute an income within the observed range. Keep units consistent and distinguish:

- predicted mean spending; and
- an individual's actual spending, which also contains residual variation.

Now you try: HouseSize regression

Time allowed: 10 minutes

Regress market price (AUD thousands) on house age.

1. Write and interpret the fitted equation.
2. Interpret the slope in dollars per year.
3. Predict the change associated with five additional years of age.
4. Interpret R^2 and the slope p-value.
5. Compare fitted and observed values for one house.

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Residual plots

A useful linear model has residuals that are:

- centred around zero;
- randomly scattered rather than curved;
- roughly constant in vertical spread; and
- not dominated by a few influential cases.

A pattern in residuals is information the model failed to capture.

Watch me: inspect residuals

Create predicted values and residuals for the age model.

1. Plot residuals against fitted values.
2. Look for curvature or a funnel shape.
3. Identify unusually large residuals.
4. Return to the original data before deciding whether any point is erroneous.

Now you try: model audit

Time allowed: 7 minutes

Audit the IncomeSpend model:

1. inspect its residual plot;
2. identify one strength and one limitation;
3. assess whether the line is suitable for prediction;
4. name one omitted variable; and
5. state whether a causal interpretation is justified.

Extrapolation

Prediction outside the observed X range is extrapolation.

- The relationship may change beyond the data.
- A mathematically valid calculation can be substantively implausible.
- Report whether a requested X is inside the sample range.
- Prefer collecting relevant data to extending the line blindly.

Now you try: prediction guardrails

Time allowed: 5 minutes

Before predicting spending for a household income of AUD 500,000:

1. check the observed income range;
2. label the request interpolation or extrapolation;
3. explain the main risk;
4. recommend an alternative if the request is unsupported.

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Watch me: a defensible regression brief

A concise brief contains:

1. business question and variable roles;
2. scatterplot and correlation;
3. fitted equation with coefficient interpretations;
4. R^2, slope test, and prediction;
5. residual and range checks; and
6. a recommendation with limitations.

Now you try: choose the better predictor

Time allowed: 10 minutes

Using **HouseSize**, compare separate simple regressions of market price on:

- house age; and
- house size.

Recommend the more useful single predictor using plots, correlation, R^2, slope evidence, residuals, and business reasoning. Explain why this is not yet a multiple-regression conclusion.

Common traps

- Reversing X and Y.
- Reporting correlation without examining the scatterplot.
- Interpreting the intercept when $X = 0$ is meaningless.
- Omitting units or “on average” from a slope interpretation.
- Claiming R^2 measures causality or accuracy.
- Extrapolating without checking the observed range.
- Treating association as causal evidence.

Closing check

Can you:

- select tools for categorical and numerical relationships?
- calculate expected counts and interpret a chi-square independence test?
- interpret a scatterplot and correlation together?
- build a simple regression in Excel?
- interpret intercept, slope, R^2, and slope p-value?
- make an in-range prediction and calculate a residual?
- communicate model limitations and avoid causal overreach?